



Taif university

College of computer & information technology

Computer Engineering department

Software project for Residential Maintenance Management System

1.0 Project Selection

System Name:

Residential Maintenance Management System

System Description:

The Residential Maintenance Management System is designed to enhance the efficiency of maintenance operations within residential communities. The system enables residents to submit maintenance requests, attach visual evidence, and monitor the progress of their requests through a structured and transparent workflow.

Maintenance technicians are able to view assigned tasks, update work progress, record service details, and provide completion reports. Supervisors and property managers benefit from a centralized dashboard that presents key performance indicators, technician schedules, request statuses, and overall service quality metrics.

By improving communication between residents and maintenance personnel, the system aims to reduce response times, ensure timely resolution of issues, and raise the overall standard of facility management services in residential environments.

Main Features:

- Submission of maintenance requests with mandatory image attachments.
- Real-time tracking of request progress and completion status.
- Automated or manual technician assignment based on availability and workload.
- Administrative dashboard for monitoring all requests, performance indicators, and workflow efficiency.
- Post-service technician evaluation by residents.
- Instant notifications and status updates throughout the request lifecycle.

2.0 Stakeholder Identification**1. Residents:**

Primary end-users who submit maintenance requests, upload images, follow the status of their requests, and provide post-service ratings.

2. Technicians:

Responsible for reviewing assigned tasks, updating the status of maintenance work, and documenting the completion details of each request.

3. Maintenance Team / Supervisors:

Oversee workflow operations by assigning tasks to technicians, managing schedules, monitoring ongoing service requests, and ensuring timely responses.

4. Property Manager:

Monitors the system at a higher administrative level, evaluates service quality, reviews performance reports, and ensures operational efficiency across the residential compound.

5. System / Server:

Manages data storage, request processing, user authentication, notification distribution, and communication across all system roles.

3.0 Functional Requirements (FR)

The system will perform the following:

FR-1: Residents Management

- FR-1.1: Residents can register and create an account using their unit number and verified contact information.

- FR-1.2: Residents have the ability to view and update their personal profile.

- FR-1.3: Residents have the ability to change or reset their account password securely .

· FR-2: Maintenance Request Submission

- FR-2.1: Residents can make a new service request.

- FR-2.2: Each request must contain :

- Service Type .

- Title of the problem and full description of the issue.

- Rate of urgency .

- FR-2.3: Residents must attach multiple pictures or a short video to illustrate the problem.

- FR-2.4: All requests shall be assigned a unique tracking number and a timestamp automatically by the system.

FR-3 Status Tracking for Requests

- FR-3.1: Residents can access a live dashboard of all their requests submitted to date.

- FR-3.2: The system shall display different status for each request for example (closed ,open ,in progress)

- FR-3.3: Residents are notified automatically for any change of their request status.

FR-4: Technician Management and Scheduling

- FR-4.1: The system shall maintain a database of technicians including their skills, specialties, and availability.

- FR-4.2: The maintenance employee can assign jobs to technicians based on skill and availability.

- FR-4.3: The system shall provide a calendar view for the supervisor to schedule technicians for visit.
- FR-4.4: Technicians are able to access a dedicated interface to see their assigned jobs, update job status, and receive notifications when tasks are assigned to them .
- FR-5: Management Dashboard and Reporting
- FR-5.1: The administrator will be able to control all the requests, technicians and residents directly through one panel.
- FR-5.2: The dashboard should display the KPIs, such as:
 - Total requests opened/closed.
 - Mean time to resolution.
 - The workload and performance of technicians.
 - The most common types of requests.
- FR-5.3: Admins can also generate reports and export them.
- FR-5.4: the property manager can generate and report detailed operational reports .
- FR-5.5: The property manager can evaluate overall service quality based on consolidated resident feedback .
- FR-6: Feedback and Ratings System
- FR-6.1: When a request is marked as "Completed" the resident can rate and comment on the service .
- FR-6.2: The ratings and comments should be aggregated and available for evaluation by the system administrators and for the review of technician performance.

4.0 Non-Functional Requirement (NFR)

NFR-1: Efficiency

- NFR-1.1: The system should display any page or dashboard view in under 3 seconds for the average load.
 - NFR-1.2: At least 100 residents and 20 technicians/managers should be able to concurrently use the system with no lagging or latency .
 - NFR-1.3: Users shall be notified no later than 1 minute after an event is triggered.
- NFR-2: User convenience**

- NFR-2.1: The resident, technician and administrator user interfaces should be easy to use and to interact with
- NFR-2.2: It shall be possible to access the system from different devices using the web browsers from PCs, tablets and mobile phones.
- NFR-2.3: The process of requesting a maintenance can be no more than 5 steps long

NFR-3: Reliability and Availability

- NFR-3.1: All critical operations like requesting submission or status update should be fault tolerant recovering from errors without losing data.
- NFR-3.2: Scheduled maintenance windows should be announced to users at least 24 hours prior to the appointment.

NFR-4: Safety

- NFR-4.1: All user information, including private information of the account and images of residents shall be protected with strong encryption .
- NFR-4.2: Access limits
 - . Residents may only see their own individual requests.
 - . Technicians can only see the tasks assigned to them.
 - . Managers/admins have full access according to their permissions.
- NFR-4.3: Secure authentication login is mandatory for all users.

NFR-5: Scalability

- NFR-5.1 The system design should be able to support an increase in the number of homes without a complete redesign.
- NFR-5.2: The database must be designed to handle an increasing number of requests and media files.
- NFR-5.3: The system architecture shall utilize a dedicated media Service to handle the uploading and processing of images / videos independently from the main application server to ensure high performance.

NFR-6: Ease of Maintenance

- NFR-6.1: The system source code should be sufficiently well documented and structured so that new developers will have an easy time both modifying the system and adding new features.

4.0 UML modeling

- Use case diagram

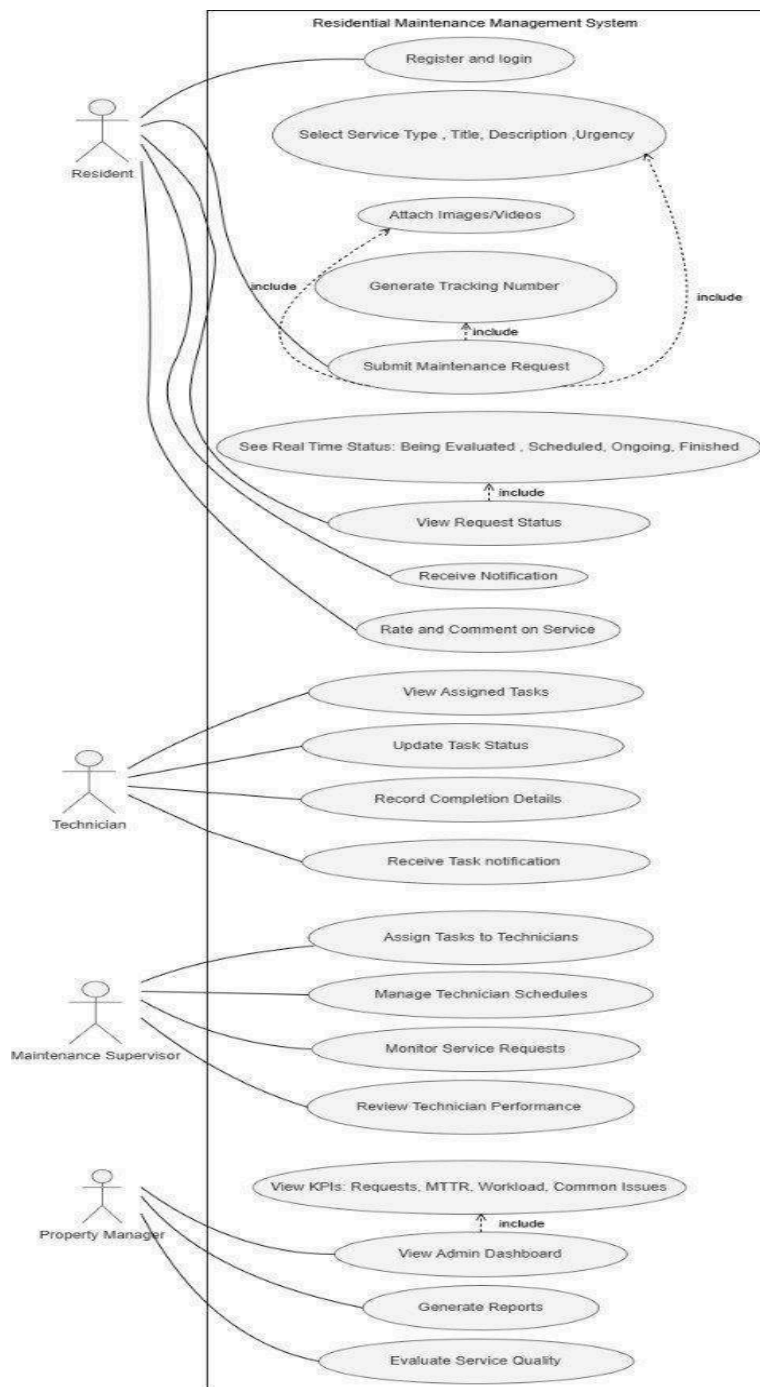


Figure 1 : Use case diagram

- Class diagram

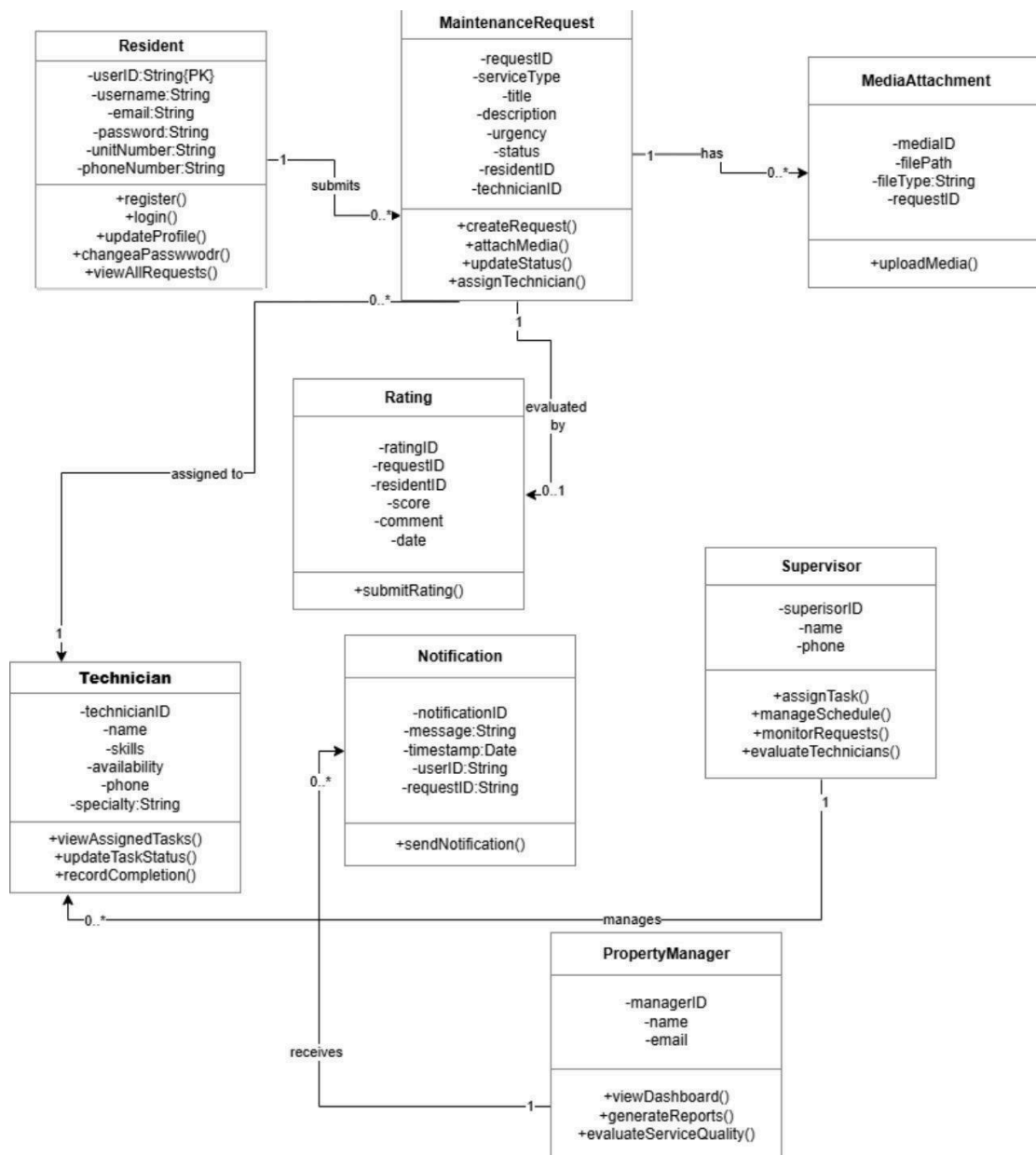


Figure 2 : Class diagram

- Sequence diagram

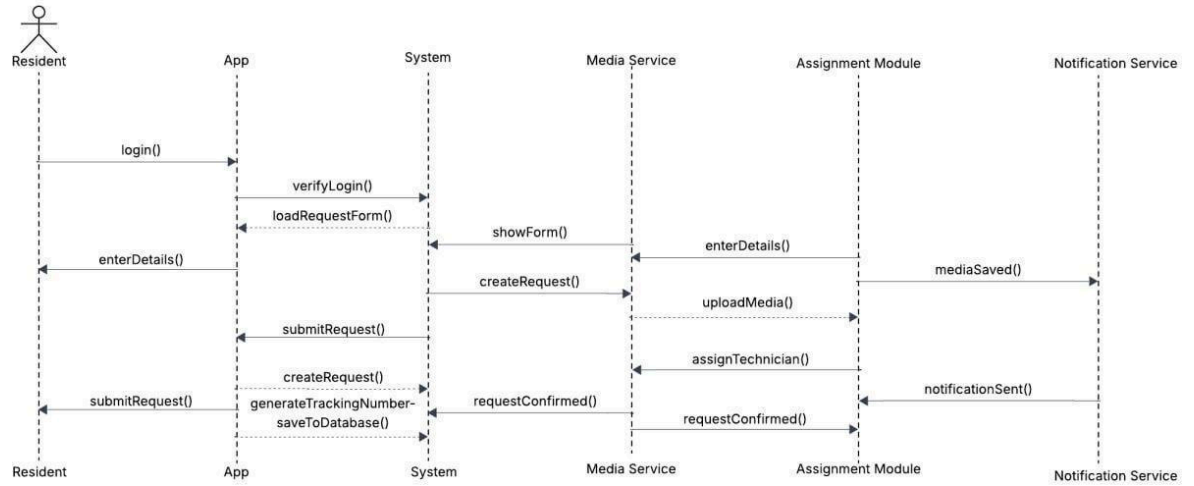


Figure 3 : Sequence diagram

5.0 System implementation and management dashboard

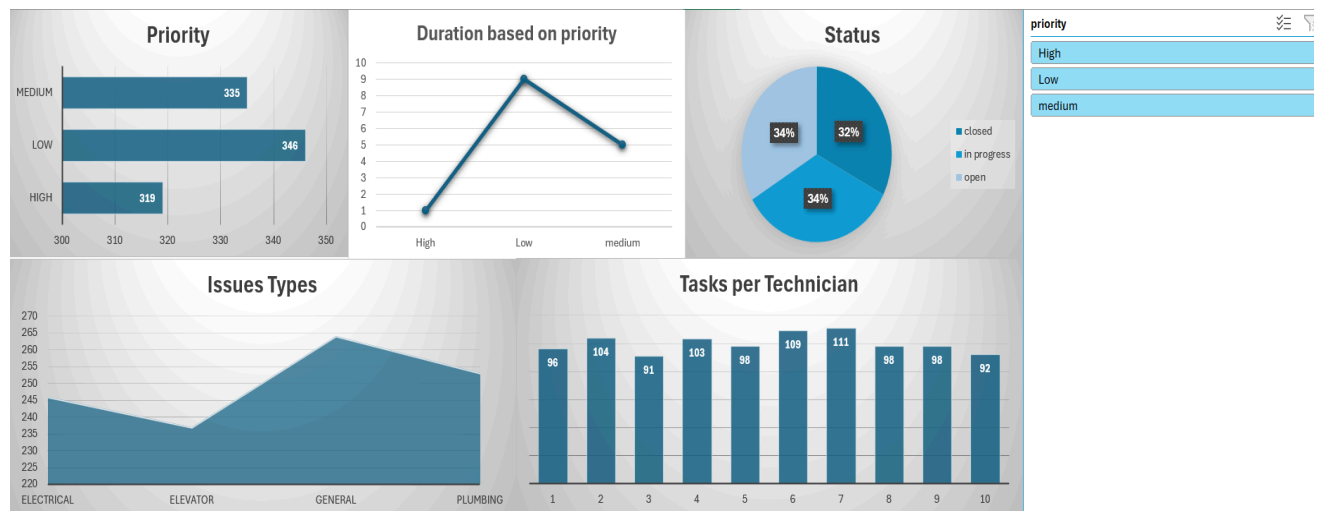


Figure 4 : Interactive property manager dashboard

This dashboard provides a comprehensive real time visual analysis of 1000 maintenance requests. It features an interactive slicer by priority that dynamically filters all associated pivot charts, Including total requests count, current task status issue types, technician workload, and average resolving time .